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IMPULSIVITY AS AN ENDOPHENOTYPE FOR SCHIZOPHRENIA: A SYSTEMATIC REVIEW

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Abstract – Research surrounding the expression of impulsivity in schizophrenia (SCZ) is extensive, as is the investigation of candidate endophenotypes for SCZ. Although much work on SCZ addresses impulsivity and its clinical manifestations, few studies propose impulsivity as an endophenotype for SCZ. This review aims to consider the evidence regarding the relationship between SCZ and impulsivity, ascertain whether implementing impulsivity as an endophenotype for SCZ is appropriate, and review current methods used when studying impulsivity utilizing the standards of the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines. A literature search was conducted using PubMed, ScienceDirect, and ProQuest. The inclusion criteria required studies to be (a) published after 2010; (b) an empirical study that collected primary data; (c) written in English; (d) published in a scholarly peer-reviewed journal; (e) conducted on Human populations; and (f) include SCZ as a primary outcome measure and discuss endophenotypes or intermediate phenotypes. Following these procedures, seven empirical studies were identified for evaluation. Results provide further evidence for impulsivity as a trait in SCZ and support its conceptualization as an endophenotype. The implications of these findings are threefold: firstly, adopting an endophenotype approach facilitates the understanding of the neurobiological substrates and genetic factors that underlie SCZ, secondly it allows for more lucid associations between genetic variants and impulsivity-related phenotypes, and lastly it recommends considering impulsivity as a construct detached from traditional diagnostic approaches.

Schizophrenia (SCZ) is a complex, chronic, polygenic, and multifactorial psychiatric disorder characterized by three subsets of symptoms: Positive symptoms (e.g., hallucinations and delusions), negative symptoms (e.g., lack of motivation, anhedonia) and cognitive symptoms (e.g., disorganized thinking, poor concentration; Green, 2016; Keefe & Fenton, 2007). Many genetic risk factors influence SCZ's clinical phenotype; in isolation, each of these genetic factors is neither necessary nor sufficient for SCZ to occur (Galindo et al., 2016). A single genetic variant can have multiple effects; genetic variants associated with SCZ risk are not specific to this disorder (i.e., pleiotropy; Meyer-Lindenberg, 2010). SCZ is a multidimensional disease involving close interactions between genetic and environmental risk factors. The interaction between these two factors during brain development can lead to deficits in neuronal synchronization and cellular damage that predispose an individual to develop the disorder (Schmidt & Mirnics, 2015).

Evidence from twin, family, and adoption studies suggests that a predisposition for SCZ is heritable (Ho et al., 2018). This has prompted extensive investigation into genetic variants that may be linked to SCZ. Notably, variations in genes such as *DISC1, CACNA1C, TCF4, and ZNF804A* have been central to these investigations (Bergen & Petryshen, 2012; Calabro et al., 2020; Harrison, 2015; Smeland & Andreassen, 2018). These genes are known for their crucial roles in neural and cortical development, calcium channel signaling, neurotransmission, transcriptional control, and synaptic processes, all of which contribute to the complexity of SCZ susceptibility (Bergen & Petryshen, 2012; Calabro et al., 2020; Harrison, 2015; Smeland & Andreassen, 2018). It is worth mentioning that numerous other risk genes and variations have been associated with SCZ, but an exhaustive list goes beyond the scope of this discussion. Although linkage studies, candidate gene association studies (Arango, 2017; Owen & Cardno, 1999), and genome-wide association studies (GWAS) have

successfully identified variants associated with SCZ (Schizophrenia Working Group of the Psychiatric Genomics, 2014), many of these results are not consistently replicable (Bergen & Petryshen, 2012). Thus, researchers may need to revise approaches by considering the compilation of many variants (i.e., single-nucleotide polymorphisms, SNPs) that might inform us about the pathogenesis of SCZ (Gualtieri, 2020; Harrison, 2015) and adjust current understandings of the phenotype.

There are several reasons to focus on the genetic mechanisms of SCZ. First, a better grasp of the biology of SCZ and the specific genetic variants underlying SCZ could better inform applications of psychiatry (Gejman et al., 2010). Such studies could allow for the development of treatments that target the biological underpinnings of the disorder rather than its symptoms (i.e., clinical manifestation) and allow treating subtypes of SCZ in an individualized way (Ozomaro et al., 2013). Though it is conceivable that further research could enable the provision of specific and tailored treatment to patients (Lally & MacCabe, 2016), the current evidence is insufficient to be applied in this way (possibly due to the heterogeneity of SCZ; Jablensky, 2006). Second, genetic studies could help reduce the stigma and misconceptions surrounding SCZ by helping people understand potential causes (Phelan, 2002). Third, genetic counseling could be improved, especially in heritability and transmissibility to offspring (Costain, Esplen, Toner, Hodgkinson, et al., 2014; Costain, Esplen, Toner, Scherer, et al., 2014). Toward this end, we posit that endophenotype approaches that identify intermediate phenotypes, symptoms, or biomarkers associated with the disorder that are more closely connected to the genetic variation are crucial in decomplexifying genetic mechanisms (Glahn et al., 2014). An endophenotype is a specific and measurable trait that provides a link between the genetic factors of a condition and its observable clinical symptoms (Gottesman & Gould, 2003). Considering consistent issues with poor phenotype measurement in behavior genetics research that makes the subtle, context-dependent effects even harder to identify or replicate, a focus on appropriate endophenotypes is a crucial step in this line of research (Glahn et al., 2014).

The literature on impulsivity in SCZ and related psychiatric disorders is extensive, as are the publications mentioning candidate endophenotypes in SCZ. Nonetheless, very few reviews define impulsivity as an endophenotype for SCZ. Much of the impulsivity literature has focused on themes, including its conceptualization – and the challenges surrounding its definition –, different assessment methods, correlations with substance use, suicidal risk, aggression, and the neural substrates involved (Ouzir, 2013). Although to our knowledge no existing theorists argue against the use of impulsivity as an endophenotype for schizophrenia, there has not been a thorough review to ensure that it does meet these criteria.

Of the articles that review impulsivity as an endophenotype, many do so in conjunction with related disorders such as attention deficit hyperactivity disorder (ADHD; Glahn et al., 2014; Robbins et al., 2012), obsessive-compulsive disorder (OCD; Fineberg et al., 2010; Robbins et al., 2012), substance abuse disorders (Jupp & Dalley, 2014), compulsive eating disorders (Howard et al., 2020), bipolar disorder (BD; Sanches et al., 2014; Wessa et al., 2015), and autism spectrum disorder (ASD; Fineberg et al., 2010). This approach provides relevant information, such as shared phenotypes, but inevitably constricts our understanding of processes specific to SCZ (Robbins et al., 2012). Viewing impulsivity through a transdiagnostic perspective, meaning the examination of impulsivity across a range of psychiatric disorders rather than within a singular disorder, holds promise for identifying common traits among these conditions and for discovering potential pathways to novel therapeutic (Fortgang et al., 2016; Robbins et al., 2012).

Current Research on Impulsivity and the Genetics of SCZ

Although genetic studies sometimes report associations between impulsivity and specific gene variants, this information is not typically discussed in the context of a given pathology and its clinical expressions. Indeed, very few recent studies assess those variables in a shared genetic and clinical framework for SCZ. Nonetheless, genetic analyses that measure impulsivity in SCZ report the involvement of distinct variations in SNPs.

Of the few studies connecting SCZ, impulsivity and genetic variation, genetic variants associated with serotonin and calcium channel functioning have been investigated. Genetic variation in the serotonin$_{2a}$ (5-HT2A) receptor, which plays a role in sustained attention (Vollenweider et al., 2007), particularly impulsivity (Quednow et al., 2012), were associated with impulse control assessed with a subset of the Continuous Test Performance (CPT) in patients with SCZ but not in healthy controls (HC; Tsuang et al., 2013). Genetic variation in a calcium channel, the A allele of *CACNA1C* SNP rs11062296 is associated with lower scores on the impulsiveness sub-measure of the PANSS (Positive and Negative Syndrome Scale; Calabro et al., 2020).

The increasing use of whole-genome sequencing has also allowed researchers to adopt reverse genetics, which involves working from the understanding of a gene to trace it back to the protein and understand its role in the cell (Takahashi et al., 1994). Coupled with imaging genetics procedures, using reverse genetics in HC uncovered an association between a risk variant in *AMBRA1* (rs11819869) and impulsivity-related traits (measured by BOLD responses during a Stop Signal Task on motor inhibition; Heinrich et al., 2013). Despite several relevant studies investigating associations between impulsivity and genetics relevant to SCZ, there is a lack of a consistent theoretical understanding of whether impulsivity functions as an endophenotype for SCZ.

Importance of Endophenotypes in SCZ Research

One of the main difficulties of genetic research in SCZ lies in the weak correlation between the clinically observable phenotype (SCZ symptoms) and the genotype (Ouzir, 2013). Indeed, some types of SCZ are not associated with the genes studied, and unaffected individuals – especially within families where SCZ is present – may present genetic susceptibility factors without observable clinical expression. Though some studies consider impulsivity, genetics, and SCZ within the same article, they do not incorporate impulsivity as an endophenotype. Novel approaches have been developed to identify subclinical traits (or endophenotypes) in patients and their relatives. Thus, by studying these more elementary and measurable phenotypes, they attempt to decrease the genetic complexity of the disease (Calkins et al., 2007).

Such concepts must satisfy the following criteria in order to meet the definition of endophenotype (or intermediate phenotype): (a) They must be associated with the disease; (b) they must be heritable; (c) they must co-segregate with the illness within families; and (d) they must be independent of the clinical state (Gottesman & Gould, 2003).

Conceptualization and Measurement of Impulsivity

Impulsivity is commonly described as following one's inner instincts and reacting to external stimuli without considering the implications and consequences that might result from that action (Moeller et al., 2001). Clinically, the DSM-5 further implies that impulsive decisions can harm individuals and bear on clinical outcomes (American Psychiatric Association, 2013; Congdon & Canli, 2008; Kaladjian et al., 2011). Impulsive behaviors are regularly associated with hazardous decision-making outcomes and adverse response inhibition (Courtney et al., 2012; Moeller et al., 2001), violence and aggression (Perroud et al., 2011; Volavka & Citrome, 2008), suicidal tendencies (Gut-Fayand et al., 2001), and substance abuse (Dervaux et al., 2010; Gut-Fayand et al., 2001).

Within a laboratory setting, impulsivity is typically measured either by means of self-report questionnaires or through experimental tasks (Ouzir, 2013). The former probes participants to rate the extent to which their behavior reflects impulsive tendencies in everyday life (e.g., the BIS-11, UPPS; Patton et al., 1995; Reddy et al., 2014), while the latter evaluates participants' aptitude to inhibit motor responses (e.g., CPT, BART, Go/NoGo, SKIP, Stop-Signal; Chamberlain & Sahakian, 2007; Logan et al., 1997; Moeller et al., 2001). Among the varying forms of impulsivity identified, two main categories emerge: a motor – or action – impulsivity and a cognitive – or choice – impulsivity (Gray et al., 2018). Motor impulsivity is expressed following a spontaneous reaction without prior reflection and motor inhibition, whereas cognitive impulsivity is characterized by a lack of evaluation of a given decision's immediate and long-term (primarily negative) consequences (Caswell et al., 2015; Ouzir, 2013). In this regard, several neurological and psychiatric conditions challenge the notion of impulsivity as an umbrella concept.

SCZ is paired with deficits in response inhibition (Enticott et al., 2008; Kaladjian et al., 2011; Nolan et al., 2011) and high scores of self-reported impulsivity compared to HC (e.g., on the BIS-11; Kaladjian et al., 2011; Nolan et al., 2011). Nonetheless, self-report questionnaires and experimental tasks assessing impulsivity have a very low correlation (Caswell et al., 2015; Enticott et al., 2008; Nolan et al., 2011; Reddy et al., 2014), and these conclusions are not consistently reached in other studies which fail to convey increased self-reported impulsivity levels (Reddy et al., 2014). Both aspects thus maintain the multidimensionality of impulsivity as a trait and support that different subtypes of impulsivity are associated with distinct but shared neural mechanisms (Ho et al., 2018).

Heritability of Impulsivity

Impulsivity is heritable, with twin studies estimating a heritability between 38-60%. (Fortgang et al., 2016). Unaffected adult first-degree relatives (FDR) of individuals with SCZ present higher levels of impulsivity on self-report and behavioral tasks (more inclined to immediate rewards on delay-discounting tasks) in comparison to unaffected second-degree relatives (SDR) and adolescents with no family history of SCZ (Ho et al., 2018). FDR's performance during delay-discounting also significantly correlated with alterations in brain regions

correlated with impulsive decision-making in SCZ individuals (Barry et al., 2020). Conversely, SDRs perform similarly to adolescents with no family history of SCZ both on self-report and behavioral tasks. Seven-year-old children of individuals with SCZ also show no difference in performance compared to matched HC (Hemager et al., 2021). The presence of impairments on measures of impulsivity in adult FDRs as opposed to children supports a developmental explanation rather than the outright rejection of impulsivity as an endophenotype and risk marker of SCZ because it potentially implies that such impairments will develop at a later age (Hemager et al., 2021). The heritability and stability of impulsivity are widely supported across the literature (Bevilacqua & Goldman, 2013; Congdon & Canli, 2008; Fortgang et al., 2016; Ho et al., 2018; Pedersen et al., 1988).

The mechanism of the heritability of SCZ and impulsivity can be further understood by identifying the biological differences associated with these phenotypes that may be affected by genetic variation. Impulsivity is associated with differences in the inferior frontal gyrus, dorsolateral prefrontal region, superior medial frontal region, pre-supplementary motor area, subthalamic nucleus, globus pallidus, striatum, and thalamus (Aron et al., 2007; Barkataki et al., 2008; Arce et al., 2006; Nolan et al., 2011; Stuss et al., 2003), which are also implicated in impulsivity in disorders including SCZ (Kaladjian et al., 2011). Because SCZ and impulsivity are highly heritable, highly associated with one another, and share related biological mechanisms, impulsivity is a good candidate endophenotype for SCZ (Barry et al., 2020). Findings from studies on unaffected relatives of SCZ patients similarly endorse impulsive behaviors as an endophenotype for SCZ in children, adolescents, and adults (Ho et al., 2018).

Impulsivity as a Non-Unitary Construct

The overwhelming majority of studies assessing impulsivity or impulse control in patients with SCZ or schizoaffective disorders allude to the discrepancies in a clear and unique definition of impulsivity and the disparateness of deficits or lack of in those patient populations (Strickland & Johnson, 2021). The divergence between those studies calls for a better delimitation of the concept of impulsivity and more consistent use of self-report and behavioral tasks in patient evaluations.

The variations and discrepancies between the conceptions of impulsivity as an umbrella term versus impulsivity characterized by distinguished subtypes arise from non-consistent methods of neuropsychological test measurements. Despite the growing body of research inquiring into different subtypes of impulsivity, much debate persists concerning the conceptualization and assessment methods of this psychological construct (Strickland & Johnson, 2021). Although impulsivity is one of the most frequently listed diagnostic criteria in the DSM (Johnson et al., 2017; Smith et al., 2007; Whiteside & Lynam, 2001), its multidimensionality is still poorly understood. Considering impulsivity as an entity and a construct not only allows for a dissolution of the ambiguity but also expands the possibilities of studying impulsive behavior subtypes individually and independently from one another.

Although experimental tasks aid in comparing the effects and patterns of one type of impulsive behavior in a population, a failure to consider their respective psychometric properties further fosters confusion about the underlying mechanisms of impulsive behaviors and the reliability and validity of findings in a real-world setting or an alternative population (Enkavi et al., 2019).

The aims of this review are: (a) To review the evidence discussing impulsivity an endophenotypes with SCZ as a primary outcome, (b) evaluate measurement and understanding of the impulsivity construct, and (c) to determine whether the implementation of impulsivity as an endophenotype for SCZ is appropriate. Although much of the literature on SCZ addresses impulsivity and its clinical manifestations, few studies have sought to propose impulsivity as an endophenotype for SCZ. This review thus aims to review the evidence discussing impulsivity as an endophenotype for SCZ to evaluate trends in current research and establish the need for a shift in theoretical framework regarding the genetics of SCZ.

Methods

The reporting of this systematic review was guided by the standards of the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) Statement (Page et al., 2021).

Eligibility Criteria

The inclusion criteria were (a) publication date after 2010; (b) being an empirical study that collected primary data; (c) written in English; (d) being published in a scholarly peer-reviewed journal; (e) conducted on Human populations; and (f) include SCZ as a primary outcome measure and discuss endophenotypes or intermediate phenotypes. Lastly, studies were excluded from the review if they were unpublished thesis and dissertation studies.

Information Sources and Search Strategy

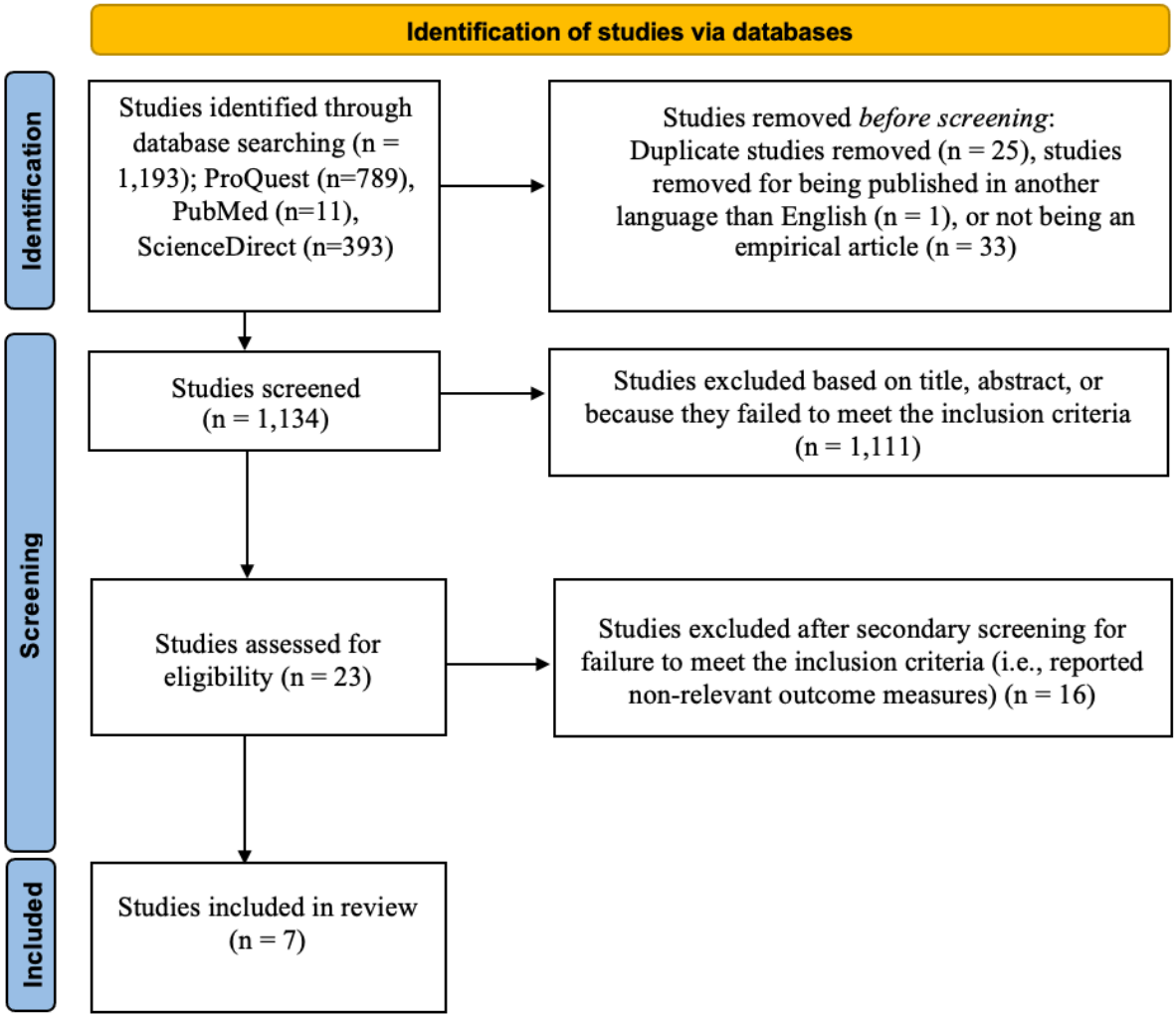
A search of the literature was conducted using PubMed, ScienceDirect, and ProQuest (which included

the following databases: APA PsycArticles, APA PsycInfo, and Publicly Available Content Database). Internet searches on those databases were conducted in March 2022 using the following search terms: (Schizophrenia) AND (impulsivity OR impulsive behavior OR impulsive behaviour OR disinhibition) AND (endophenotype OR intermediate phenotype) AND (genetics OR genetic). After an additional hand search, documents that did not fit the desired study type (e.g., poster presentations, conferences, e-posters, or oral communications) were excluded. Moreover, studies that (a) did not include SCZ as a primary outcome measure or (b) did not assess impulsivity or endophenotypes were excluded.

Study Selection and Data Collection Processes

Based on the results obtained after the initial search, the title and abstract of each study were screened, and one reviewer further evaluated potentially relevant studies for eligibility. Studies were included if they had a mention of "impulsivity" or related terms and, when possible, included the term "endophenotype" or "intermediate phenotype." The information on study selection and eligibility assessment can be found in the PRISMA flow diagram (Figure 1). Among the 1,193 results obtained through database searching, 59 studies were removed due to duplication (n = 25), publication in another language than English (n = 1), or not being an empirical research article (n = 33). After the title and abstract of each study were examined, 1,111 studies were excluded because of unsuitability for the present review. A secondary screening procedure was done to reassess the relevance of the 23 eligible studies. After reviewing the full text of those articles, 16 were excluded because their focus and outcome measures emphasized impertinent traits or disorders. Following these procedures, seven empirical studies were evaluated in the present review.

Figure 1
PRISMA Flow Diagram of the Studies Selection Process



Results

The analysis of the findings from the seven articles included revolves around three core aims: first, an exploration of the various dimensions of impulsivity and their measurement; second, an investigation into methodological variations, primarily related to the assessment of impulsivity; and finally, an evaluation of impulsivity as a potential endophenotype for SCZ within the context of established criteria.

Study characteristics

Characteristics of the studies included in this review can be found in Table 1.

Methodological variations

Participants

The seven studies included in this review include 1,036 participants, with studies ranging in sample size from 60 participants to 420 participants. In total, there were 310 participants with a SCZ or schizoaffective disorder diagnosis, and 200 HCs. One of the aims of this subsection was to state the gender distribution within the selected articles. Most of the studies reviewed enrolled more male participants (n = 573; 55.31%) than female participants (n = 463; 44.69%). Except for Ho and colleagues' study design (2018), which did not include individuals with SCZ, every other study enrolled participants who were medicated, primarily using neuroleptics (Duva et al., 2011; Fortgang et al., 2016; Hoptman et al., 2014; Kaladjian et al., 2011; Nolan et al., 2011; Reddy et al., 2014). Two studies were designed and conducted outside the United States; Fortgang and colleagues (2016) carried out their research in Sweden, and Kaladjian and colleagues (2011) conducted their study in France. Because all the studies reviewed here support the presence of impulsive tendencies in their participants – albeit to different extents –we were unable to identify cross-cultural differences in expression of impulsivity in SCZ cohorts.

Medication

Of the studies reviewed here, six included patients with SCZ on AP medication (one study on the heritability of impulsivity does not include SCZ patients), and none distinguished between medicated versus non-medicated participants in their analysis. The presence of medication in participants can be responsible for the heterogeneity of results on the incidence of impulsivity in patients with SCZ (Reddy et al., 2014). Some studies show that medication use may reduce impulsive behaviors (Spivak et al., 1997), but others fail to report an association between medication and impulsivity in patients with psychosis (Ahn et al., 2011; Heerey et al., 2007; Shurman et al., 2005).

As displayed in Table 1, most participants take atypical or second-generation antipsychotics (SGAs), including Aripiprazole and Clozapine, two widely prescribed partial dopamine D2/D3 receptor agonists (Casey & Canal, 2017). Both serotonin and dopamine are neurotransmitters known to play a substantial role in impulsive behaviors and risky decision-making (i.e., risk-taking; Reddy et al., 2014; Seo et al., 2008). SGAs also reduce aggression and impulsivity in many patients (Bilder et al., 2002; Jones et al., 2011; van Schalkwyk et al., 2018; Volavka & Citrome, 2011), which prompts the investigation of the effects of dopamine stabilizers on decision making outcomes.

Measures of Impulsivity

To assess and measure impulsivity, five of the studies used the Barrat Impulsiveness Scale (BIS-11), one of the most widely used self-report questionnaires to evaluate impulsivity (Duva et al., 2011; Fortgang et al., 2016; Kaladjian et al., 2011; Nolan et al., 2011; Reddy et al., 2014). The two other studies gave participants the UPPS, an equivalently widely used self-report scale (Ho et al., 2018; Hoptman et al., 2014). Both Hoptman and colleagues (2014) and Nolan and colleagues (2011) administered additional self-report questionnaires, the Buss Perry Aggression Questionnaire (BPAQ) and the Life History of Aggression (LHA). Hoptman and colleagues (2014) also gave their participants the Impulsive and Premeditated Aggression Scale (IPAS); it is the only study to have used three different self-report measures. Regarding experimental tasks, four studies used the SST and Go/No-go task, reaction time tests that precisely measure response inhibition (Duva et al., 2011; Fortgang et al., 2016; Kaladjian et al., 2011; Nolan et al., 2011). Two studies performed the BART, a paradigm that measures the degree of risk-taking behaviors (Duva et al., 2011; Reddy et al., 2014). Finally, two studies chose to administer the SKIP, TCIP (both modifier versions of the latter test), and the Delay Discounting Task (Duva et al., 2011; Ho et al., 2018).

Evidence Relating to Endophenotype Criteria

The following section reviews the main findings from our studies in relation to how they apply to the outlined criteria for an endophenotype.

Criterion 1: Association with Illness

To substantiate the association between impulsivity and SCZ, a consistent pattern of findings has emerged across several studies. Self-report measures consistently indicated heightened impulsivity and aggression scores in individuals with SCZ compared to HC and proband groups (with p values ranging from .01

Table 1
Main Characteristics of the Studies Reviewed (N=7)

Study	Participants	Impulsivity measures	Medication	Study Location
(Fortgang et al., 2016)	Participants with SCZ (n=64), SCZ Co-twins (n=64), BD*** (n=71), BD Co-twins (n=51), MDD*** (n=44), MDD Co-twins (n=25), Other psychiatric illness (n=5), Other Co-twins (n=2), No consensus diagnosis (n=33), HC (n=61)	BIS-11, SSS, SST	AP medication (n=62) Mood-stabilizers (n=44), Antidepressants (n=71)	Sweden
(Hoptman et al., 2014)	Participants with SCZ (n=27), schizoaffective disorder (n=6), HC (n=31)	UPPS, BPAQ*, IPAS*, LHA*	Clozapine (n=6)** Mood-stabilizers (n=12)	USA
(Kaladjian et al., 2011)	Participants with SCZ (n=26), HC (n=30)	BIS-11, Go/NoGo	All participants were taking psychotropic medication Atypical AP (n=21) Typical AP (n=5)	France
(Reddy et al., 2014)	Participants with SCZ (n=38), BD*** (n=68), HC (n=36)	BIS-11, BART	SCZ participants taking AP medication (n=37)	USA
(Duva, Silverstein, & Spiga, 2011)	Participants with SCZ or schizoaffective disorder (n=49), psychosis NOS (n=11)	BIS-11, BART, Laboratory behavioral measures of impulsivity (includes GoStop Impulsivity Task, SKIP*, TIME*, TCIP*)	All participants were taking AP medication.	USA
(Nolan et al., 2011)	Participants with SCZ or schizoaffective disorder (n=42), HC (n=42)	BIS-11, BPAQ, LHA, SST*	Treatment with either second-generation neuroleptics (n=28), or with a combination of first- and second-generation neuroleptics (n=11(missing data for n=3))	USA
(Ho et al., 2018)	First-degree relatives of SCZ patients (n=39), second-degree relatives of SCZ patients (n=53), adolescent controls with no SCZ family history (n=118)	UPPS-P, Delay Discounting Task	N/A	USA

* BPAQ = Buss Perry Aggression Questionnaire, IPAS = Impulsive and Premeditated Aggression Scale, LHA = Life History of Aggression, SKIP = Single Key Impulsivity Paradigm, TIME = Time Estimation Task, TCIP = Two Choice Impulsivity Paradigm, SST = Stop-Signal Task
** Medication was converted to chlorpromazine equivalents. One participant was taking asenapine
*** BD = Bipolar Disorder, MDD = Major Depressive Disorder

to less than .001 and specific statistical values reported in study descriptions below; Duva et al., 2011; Fortgang et al., 2016; Ho et al., 2018; Hoptman et al., 2014; Kaladjian et al., 2011; Nolan et al., 2011; Reddy et al., 2014).

In Nolan and colleagues' study (2011), patients exhibited longer Stop Signal Reaction Time (SSRT) — a behavioral measure of motor inhibition — and increased response variability, with $t(82) = 4.84$, $p < .001$. Risk-taking behaviors were explored in two studies, both indicating heightened impulsivity and risk-aversion in individuals with SCZ (Duva et al., 2011; Reddy et al., 2014). Duva et al. (2011) found a significant main group effect on the TCIP task, revealing that individuals with comorbid psychotic disorders (SCZ, schizoaffective disorder, or psychosis not otherwise specified) and current or past substance abuse demonstrated a heightened propensity for immediate gains compared to HC ($F(2, 57) = 4.60$, $p = 0.01$). In addition, two investigations into the neural basis of impulsivity in SCZ underscore the involvement of prefrontal brain regions, as evidenced by Hoptman (2014) and Kaladjian (2011). Kaladjian and colleagues (2011) unveiled a positive correlation in patients between BIS-11 scores and cerebral activation related to response inhibition, particularly within the right ventrolateral prefrontal cortex ($r = 0.84$, $p < .001$), compared to HC ($r = 0.20$, $p = .293$). Hoptman and colleagues (2014) found that the positive ($d = 1.50$) and negative ($d = 1.21$) urgency subscales were higher in patients than HC (with $p < .001$), whereas lack of premeditation, lack of perseverance and sensation seeking were not (Group×Subscale Interaction, $F(4,60) = 14.86$, $p = 1.05\times10^{-7}$). Further, they identified a robust association between cortical thinning in ventral prefrontal regions and increased impulsivity in SCZ patients. Specifically, they found an association between negative urgency and reduced cortical thickness in the right frontal pole ($r(30) = -0.64$, $p < .001$) and right medial orbitofrontal cortex ($r(30) = -0.36$, $p = .042$). Positive urgency was also negatively correlated with thickness in the left rostral anterior cingulate ($r(30) = -0.38$, $p = .031$) and the right frontal pole ($r(30) = -0.44$, $p = .013$).These results imply a connection between impulsivity and reduced efficiency in processing inhibitory control within the prefrontal brain regions.

Criterion 2: Heritability

Fortgang and colleagues (2016) examined the overlap and genetic loading of impulsivity across SCZ, BD, and major depressive disorder (MDD). Their best fitting models indicate that genetic factors explain between 38%-60% in the variance across various subsets of impulsivity, including self-reported attentional, motor, and non-planning impulsivity, self-reported sensation seeking, and SSRT. SCZ patients exhibited longer SSRTs than HC ($d = 0.78$, $p = .010$), and their co-twins ($d = 0.46$, $p = .004$), aligning with the results observed in Nolan and colleagues' study.

Criterion 3: Co-segregation within Families

Ho et colleagues (2018) contributed to our understanding of heritability by measuring impulsivity in unaffected biological relatives of SCZ patients. They found that on a delay-discounting task, first-degree relatives of SCZ patients exhibited increased impulsivity-related behaviors and displayed a greater preference for immediate rewards compared to adolescents with no SCZ family history ($Q \geq 3.13$, $p \leq .006$) and second-degree relatives ($Q \geq 2.40$, $p \leq .04$). This finding suggested a positive association between impulsivity and SCZ risk, with the severity of impulsivity increasing with closer familial relatedness to the SCZ proband.

Criterion 4: Independence from Clinical State

Impulsivity is a prevalent trait that is observed in both clinical and non-clinical populations, indicating that it is a widespread human characteristic (Chamorro et al., 2012). An overview of the total mean BIS-11 scores obtained by HC within the studies included in this review demonstrates that even within non-clinical groups, moderate levels of impulsivity are prevalent. Specifically, Kaladjian and colleagues (2011) reported a mean score of 58.9 (SD=8.6), while Reddy and colleagues (2014) reported a score of 42.9 (SD=8.8), and Nolan and colleagues (2011) reported a mean score of 56.00 (SD=9.95). It is worth noting that the BIS-11 scoring scale ranges from 30 to 120, with higher scores showing higher impulsivity. Although these values are significantly lower than those observed in patient groups, they do not negate the presence of impulsivity and emphasize that control groups exhibit impulsivity even in the absence of clinical diagnoses. This finding highlights that impulsivity is an inherent and prevalent characteristic of human behavior that is independent of clinical states.

Discussion

Although SCZ does not fall under an impulse disorder per DSM classification (American Psychiatric Association, 2013), the studies reviewed here not only provide further evidence for the presence of impulsivity as a trait in SCZ patients but support its conceptualization as an endophenotype. It is also considered an adequate intermediate phenotype in the framework of non-psychiatric populations (i.e., clinically; Chistiakov et al., 2012; Mann et al., 2009). Indeed, even though impulsivity can present stronger associations with psychiatric disorders, it remains a universal trait not

unique to clinical populations (Chamorro et al., 2012). As an endophenotype, impulsivity is thus a marker and a risk factor for SCZ as defined by the DSM-5, and as studies show, it is present throughout the spectrum. A substantial portion of the literature on SCZ has put forward endophenotype candidates that fall under distinct categories, including neuropsychology, electrophysiology, biochemistry, and neuroanatomy (Bryll et al., 2021). As per the outlined criteria for endophenotypes, this review has successfully demonstrated that impulsivity (a) is significantly associated with SCZ; (b) heritable; (c) state-independent (i.e., unaffected individuals present impulsive tendencies albeit to a lesser extent than patients); and (d) co-segregates within families of probands with SCZ. Moreover, although measures of impulsivity do not consistently bear the same outcomes, it is reliably measured (Calkins et al., 2007; Gottesman & Gould, 2003). Allowing for impulsivity to be considered an endophenotype holds many benefits, namely in expanding our understanding of the neurobiological substrates and risk factors (Ho et al., 2018), gene discovery (Bevilacqua & Goldman, 2013), diagnosis and identification of risk genes for SCZ (Radhu et al., 2017), and overall to facilitate clinical management of individuals with SCZ (Congdon & Canli, 2005; Gottesman & Gould, 2003).

Summary of Methodological Inconsistencies

The studies reviewed here support that impulsivity, in its multidimensionality, is compatible with an endophenotype definition. To reiterate, endophenotypes encompass heritable, intermediate traits that can serve as indicators for the underlying genetic basis of a complex phenotype or disorder. However, there are some inconsistencies regarding which clinical populations display impulsivity, which specific neural and molecular mechanisms are disrupted, and, how impulsivity is defined and measured (Congdon & Canli, 2005; Dang et al., 2020). These irregularities may be partially due to (a) a problem with the construct of impulsivity; (b) differing neural mechanisms involved depending on the task administered; and (c) a lack of consistency or regulation in the procedures and psychometric properties of a given test. A problem with the construct refers to the mere definition of impulsivity which encompasses many subtypes that may rely on distinct neural networks. In addition, different mechanisms might be implicated depending on the type of assessment performed; for example, the processes associated with self-perception and self-evaluation of impulsivity may differ from those engaged in task performance (Reddy et al., 2014). Lastly, it is suspected that task designs and neuroimaging procedures might contribute to the misalignments of results (Kaladjian et al., 2011). The measures reported in Table 1 illustrate the array of measurement tools used in assessing impulsivity: While some tests or paradigms measure response inhibition (e.g., SST and Go/No-go), other measures gauge the degree of risk-taking behaviors (e.g., BART). Taken together, these tasks all aim to measure impulsivity, yet the mere intricacy of this construct prevents such an overall measurement of impulsivity. Neuropsychological tasks assessing impulsivity do not systematically measure the same construct and traits under the same conditions and test methodologies. Incidentally, there appears to be a tendency in research to apply different methods (e.g., self-report and behavioral measurements) in assessing individual differences and to conflate findings across measurement types despite a weak correlation between both types of measures (Dang et al., 2020). Fortgang and colleagues (2016) highlight this belief by indicating that sensation-seeking and motor response inhibition are, in fact, detached from impulsivity, thus suggesting that phenotypes associated with impulsivity may correspond to utterly different constructs. It is thus primordial to consider impulsivity from a Research Domain Criteria (RDoC) standpoint in complement to traditional DSM-ICD diagnostic approaches (Fortgang et al., 2016; Lilienfeld & Treadway, 2016).

Toward a Better Phenotype Conceptualization

In studies investigating genetics and SCZ, there remains a tendency to use DSM criteria as a set phenotype, which inevitably results in heterogeneous findings that are not easily replicable (Braff et al., 2007; Congdon & Canli, 2005). Considering DSM criteria as an accurate and comprehensive representation of one's traits can be problematic on several levels. DSM criteria are, by definition, established as nosological guidelines that categorize a fixed group of heterogeneous symptoms for clinical purposes (Jablensky, 2010; Kendell & Jablensky, 2003). This implies that the DSM has the inherent limitations of a categorical classification system, meaning that these categories are based on syndromic groups and, thus, only on clinical criteria (Frances & Egger, 1999; Ginsburg et al., 1996; Kendell & Jablensky, 2003). The principles that prevail in the establishment of such atheoretical classifications are validity and reliability (Carson, 1991; Kendell & Jablensky, 2003). In the absence of diagnostic biomarkers that can act as external validators, the validity of the DSM (or other diagnostic tools) is hard to establish and leads to the reduction of validity to questions of reliability (Kupfer & Regier, 2011; Weickert et al., 2013).

The heterogeneity of SCZ has led to the emergence of multiple models that account for hypothetical processes (Arslan, 2018; Owen & Cardno, 1999; Smeland & Andreassen, 2018). Yet even with an abundance of genetic and clinical evidence, those different models do not reach a consensus (Peralta & Cuesta, 2001; Pickard, 2011). Relying on such categorical classifications hinders the singularity and individuality of clinical presentations (Congdon & Canli, 2005; Heckers et al., 2013). Resorting to heterogeneous symptom criteria thus also entails relying on corresponding heterogeneous genetic processes, which inevitably precludes the identification of phenotypes closer to the influence of genetic variants (Braff et al., 2007; Congdon & Canli, 2005; Gottesman & Gould, 2003). Hence, adopting an endophenotype approach that is amenable to quantitative genetic data constitutes a way to overcome this obstacle (Braff & Light, 2005; Gur et al., 2007; Hyman, 2000). Endophenotypes in their ability to identify subclinical traits and measurable phenotypes not only decrease the genetic complexity of SCZ but allow for more lucid associations between genetic variants and impulsivity-related phenotypes (De Geus & Boomsma, 2001; Gottesman & Gould, 2003; Turetsky et al., 2007).

Conclusion

Reviews comparing different neuropsychiatric disorders such as ASD, ADHD, BD, SCZ, and impulsive behavior disorders find that these conditions all share impairments in serotonin regulation, specifically in shared domains such as executive functioning, public behavior, and sensory gating (Broberg et al., 2010; Patrick & Ames, 2015). Thus, if impulsivity has been proposed as an endophenotype for those disorders (Fortgang et al., 2016; Nigg et al., 2017; Sharp, 2016; Yen et al., 2013; Yi-Ling et al., 2017; Zhou et al., 2018), we support that further investigations on SCZ alone should be carried out. Lastly, though this review supports that impulsivity should be considered an endophenotype for SCZ, it also emphasizes the need for future research – specifically genetic studies – to rely on endophenotype approaches as their central methodology.

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